GRÉGOIRE PELLETIER

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SUMMARY

Aspiring Data Scientist with experience in developing machine learning MVPs and designing AI-based thermal estimators. Skilled in using Python, Pytorch, and Tensorflow for data science projects, including statistical research and image processing. Eager to leverage academic background and practical experience to contribute to innovative data-driven solutions.

PROFESSIONAL EXPERIENCES

PARIS DIGITAL LAB Feb 2025 - Jul 2025

DATA SCIENTIST, LA DÉFENSE, FRANCE

Development of Minimum Viable Products in Machine Learning and Data Science.

- Getting SNCF's trains and station information using only LLM and Anthropic's MCP (Model Context Protocol) to easily connect consumers to internal SNCF APIs. Design of a client for secure and real-time API calls. Creation of a streamlit interface using the client to integrate LLMs' responses formatted according to SNCF's graphic charter.
- Generate and synthesize THALES intern documentation (docx, pptx) from a multimodal database (videos, PDFs, images) by unifying data in pdf format (image/video processing, contour detection...). Creation of a specialized chatbot via a SOTA multimodal RAG (ColPali) for visual retrieval in a streamlit interface.

EMOTORS Jun 2024 - Nov 2024

DATA SCIENTIST, CARRIÈRES-SOUS-POISSY, FRANCE

Internship: Design of a thermal estimator for electric machines of the STELLANTIS electric motor subsidiary using Hybrid Twin methods, RNN and adaptation of the thesis "Thermal Neural Network" (Recurrent Neural Network, TNN, Pytorch).

ENS PARIS-SACLAY May 2023 - Jul 2023

STATISTICAL RESEARCH, GIF-SUR-YVETTE, FRANCE

Internship at the Borelli Center: Use of Gaussian fields to create textures, and application to the modeling of local mammographic patches. Use of image processing and neural networks on the created data (Pytorch).

LABORATOIRE DE MATHÉMATIQUES D'ORSAY Feb 2023 - Jun 2023

SUPERVISED RESEARCH WORK, ORSAY, FRANCE

Supervised research work: "RANDOM GRAPHS - FORBIDDEN BACKWARD WALK!": Study of random graphs and use of the Hashimoto matrix (or non-backtrack matrix) for community detection in graphs.

FREELANCE 2021 - 2025

PRIVATE MATHS AND PHYSICS TUTOR, PARIS, FRANCE

Provided personalized tutoring in Mathematics and Physics to students in Paris, helping them to solidify foundational concepts, improve their understanding of complex topics, and achieve better academic results.

EDUCATION

UNIVERSITÉ PARIS SACLAY, ORSAY, FRANCE 2022 - 2026

BACHELOR'S DEGREE MAGISTERE MATHEMATICS AND MASTER'S DEGREE MATHEMATICS AND ARTIFICIAL INTELLIGENCE, Mathematics, Mathematics and Artificial Intelligence

- **Coursework:** Differential Calculus, Integration, Algebra, Measure Theory, Probability, Complex Analysis, Statistics, Statistical Modeling, Probability, Optimization, Supervised Learning, Unsupervised Learning, Machine Learning, Deep Learning, SQL, Big Data, Sequential Learning, Reinforcement Learning

LYCÉE LOUIS-LE-GRAND, PARIS, FRANCE 2019 - 2022

CLASSE PRÉPARATOIRE AUX GRANDES ECOLES, MATHÉMATIQUES ET PHYSIQUE

SKILLS AND LANGUAGES

- **Languages**: French, English, German
- **Programming**: Python, R, Spark, Ocaml, SQL, LATEX
- **Data Science & IA**: Pytorch, Tensorflow, Scikit-learn, Pandas, Git, Streamlit
- **Other**: Agile Prototyping, Design Thinking

PROJECTS

Prediction and forecasting of time series - Forecasting of net electricity demand during the "Soberty" period | https://www.kaggle.com/competitions/net-load-forecasting-during-the-soberty-period

- Data Challenge on evaluating net electricity demand by adapting to its variations during the pandemic.

The performance of chess grandmasters (Web scraping, Data exploration, Statistical learning) |

https://github.com/GregoirePelletier/Chess_games-ML

• Predict the outcome of a game between two grandmasters, knowing the ECO code of the opening, publication of the scraped dataset

Neural Stochastic Differential Equations (Deep Learning) | https://github.com/GregoirePelletier/torchsde

• Study of an article "Neural SDEs as Infinite-Dimensional GANs" and training on data from an Ornstein-Uhlenbeck process and time-dependent

TIPE: Classification des lésions en mammographie (Traitement d'image, Apprentissage automatique) | https://github.com/GregoirePelletier/Classification-de-lesions-grace-a-l-intelligence-artificielle

• Simple lesion modeling to train a K-nearest neighbors model to classify mammograms.

CERTIFICATION

Linguaskill: English, C1 or higher